

MIDDLE EAST TECHNICAL UNIVERSITY | DEPARTMENT OF ELECTRICAL AND ELECTRONICS
ENGINEERING

EE201 – Circuit Theory I

Lecture 1: Basic Concepts

Fall 2026–2027 | Section 2 | M. Mert Ankaralı

Colour code used in these notes and in the slides

■ charge q ■ current i ■ voltage v ■ power p , energy w

These notes cover the first lecture of EE201. They are meant to be read *next* to the lecture, not instead of it: the derivations are complete, but the discussion in class is where the ideas are questioned. Sections follow the order used in the recorded lecture of Prof. Emre Tuna on [METU OCW](#), so you can jump between the two:

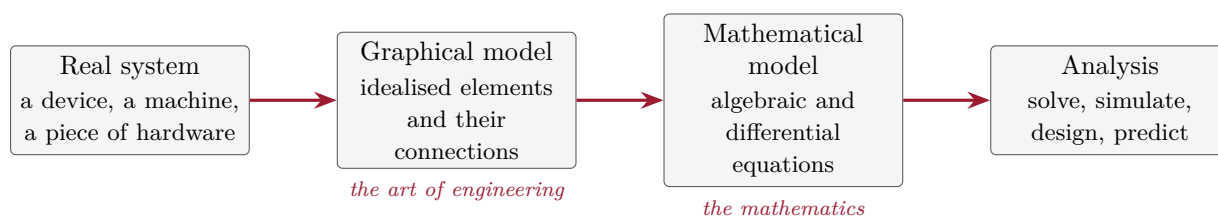
Section	Video	Section	Video
1. Why circuits? modelling	00:00	6. Power and energy	33:15
2. Charge	06:25	7. Passive sign convention	36:07
3. Current	13:12	8. Circuits, lumped circuits	48:46
4. Voltage	21:21	9. Terminals and nodes	51:00
5. Ground and node voltages	28:40		

The complete set of course notes by Prof. Emre Tuna is at users.metu.edu.tr/etuna/ee201.

1. Why circuits? From the world to a model

Circuit theory is not really about wires. It is about a particular, extremely successful way of *modelling* physical systems, and you will use the same way of thinking in control, signal processing, power, robotics and electronics for the rest of your career.

The pipeline is always the same three steps:



The hard step is the first arrow. A real resistor has inductance, a temperature coefficient, a finite power rating and a physical size; we throw all of that away and keep one number, R . Whether that is brilliant or foolish depends entirely on what you are asking of the model — and knowing the difference is what makes an engineer.

“All models are wrong, but some are useful.”

George E. P. Box, *Science and Statistics*, 1976

Keep that sentence in mind for the whole semester. Every element you will meet in EE201 is wrong in some measurable way. The engineering question is never “is this model true?” but “is it useful for what I am about to do with it?”.

The same pipeline, other domains

The elements change, the structure does not. Every such domain has one *across* quantity and one *through* quantity:

Domain	“Across” quantity	“Through” quantity
electrical	electric potential	current
hydraulic	pressure	volumetric flow rate
thermal	temperature	heat flow
mechanical	velocity	force

An across quantity is always a *difference* between two points, in every one of these domains — and in every one of them the word “difference” is usually dropped. We say *voltage* rather than “potential difference”; a plumber says “pressure” rather than “pressure difference”. The difference has not gone away; it is embedded in the name, and the two points are understood from the context. Each of these domains obeys the same conservation and connection laws you will learn here, which is why circuit theory is taught to *all* engineers and not only to electrical ones.

In EE201 we work with the *lumped circuit* model, which we make precise in Section 8. The first seven sections build the vocabulary that model is written in: charge, current, voltage, ground, power, and a sign convention that keeps the bookkeeping honest.

2. Charge

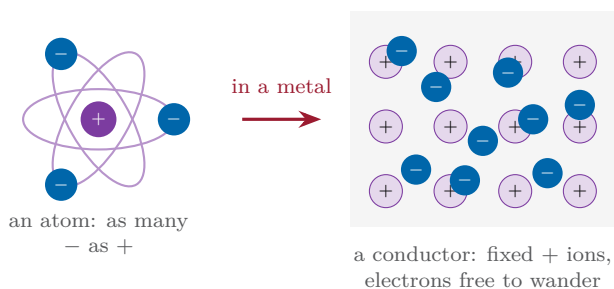
Electric charge

Charge is the physical property of matter that makes it experience a force in an electromagnetic field. It is denoted by q (or Q for a constant charge) and measured in **coulombs** (C).

Charge is quantised: it comes in integer multiples of the elementary charge

$$e = 1.602 \times 10^{-19} \text{ C}, \quad \text{so} \quad 1 \text{ C} \approx 6.24 \times 10^{18} \text{ electron charges.}$$

An electron carries $-e$ and a proton $+e$. In an isolated atom the two exactly balance, so the atom as a whole is neutral.



Two facts about charge do all the work in this course:

1. **Charge is conserved.** It is neither created nor destroyed; it only moves. When we write Kirchhoff’s current law in Lecture 2, we are writing conservation of charge.
2. **A conductor is already full of charge, and is still neutral.** In a metal, roughly one electron per atom breaks loose and is free to move through the material, leaving a fixed positive ion behind. One cubic centimetre of copper contains about 8.5×10^{22} such free electrons —

about 13 600 C of *mobile* charge — while the net charge of that copper is essentially zero, because every free electron is matched by the ion it left.

The second fact is easy to misread, so it is worth stating the consequence explicitly: a current in a wire is not charge being *poured into* the wire. The charge is already there. A current is that charge *drifting*, and while it drifts the wire stays neutral. This is exactly assumption (a) of the lumped model in Section 8, and it is what makes Kirchhoff’s current law true.

3. Current

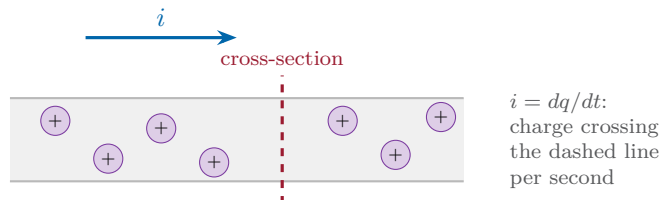
If charge sits still, nothing interesting happens. Circuits are about charge *in motion*.

Electric current

Current is the time rate of flow of charge through a cross-section:

$$i(t) = \frac{dq(t)}{dt} \iff q(t) = q(t_0) + \int_{t_0}^t i(\tau) d\tau.$$

The unit is the **ampere (A)**, and $1 \text{ A} = 1 \text{ C/s}$.



Example 1.1 — a coulomb, and the speed of an electron

(a) **How much charge does a phone battery hold?** A 4000 mAh cell:

$$Q = 4 \text{ A} \times 3600 \text{ s} = 14\,400 \text{ C}, \quad N = \frac{14\,400}{1.602 \times 10^{-19}} \approx 9 \times 10^{22} \text{ electrons.}$$

Note first that the rating is given in ampere-hours — a charge, written as a current multiplied by a time. Note second what 10^{22} means for us: charge really does come in indivisible lumps of e , but at this scale the lumpiness is utterly invisible, in the same way that a litre of water is not visibly made of molecules. That is the licence we need to treat $q(t)$ as a smooth function of time and to write dq/dt at all.

(b) **How fast do the electrons actually move?** Take 1 A flowing in an ordinary 1 mm² copper wire. Each electron drifts along the wire at roughly

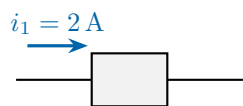
$$0.07 \text{ mm/s} \quad \text{— about one metre in four hours.}$$

Yet the lamp lights the instant you flip the switch. The carriers crawl; the *signal* that sets them all moving travels at nearly the speed of light. Keep that gap in mind — it is precisely what Section 8 turns into the condition $d \ll \lambda$.

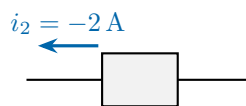
(For the curious: the drift speed follows from $v = i/(n e A)$, where $n \approx 8.5 \times 10^{28} \text{ m}^{-3}$ is the density of free electrons in copper. You will not need this formula in EE201 — it is here so that the number is not magic.)

3.1. Reference direction: the arrow is a choice, not a fact

A current value is meaningless without an arrow. The arrow declares *which way we call positive*; the sign of the number then tells us which way the charge actually moves.



charge flows to the right



exactly the same physical situation

Both pictures describe one and the same physical current. In circuit analysis you will often assign arrows before you know the answer; a negative result is not a mistake, it is the model telling you the charge goes the other way.

Conventional current

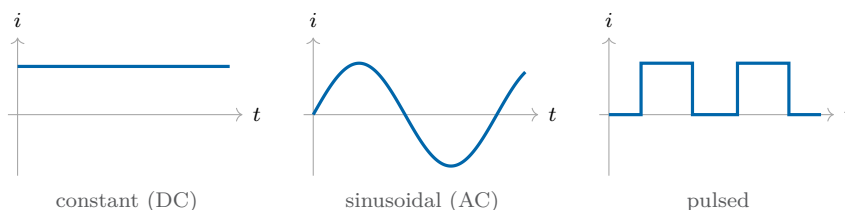
By historical convention, the current arrow points in the direction in which *positive* charge would move. In a copper wire the carriers are electrons, so they actually move *against* the arrow.

This changes no equation in this course, and it is worth being clear about why. A current counts only the *net rate at which charge crosses a section*: one electron ($-e$) moving to the left transports exactly as much charge to the right as one proton ($+e$) moving to the right. The model never asks what the carriers are — and it had better not, because in a semiconductor they are electrons *and* holes, and in a nerve cell or a battery they are ions. The convention is older than the discovery of the electron, and we keep it because nothing depends on it.

3.2. Notation

We follow the usual convention of the course: lowercase letters for quantities that may vary with time, uppercase for constants.

$i(t), v(t), q(t), p(t)$ (general) I, V, Q, P (constant, “DC”)



Only the first of these deserves the symbol I ; the other two must be written $i(t)$. Most of EE201 is about circuits whose currents and voltages change in time, so get used to the lowercase notation early.

Example 1.2 — charging a phone

In the last part of a charge cycle the battery is topped up: the charge *stored in the battery* approaches its full value as

$$q(t) = Q_f (1 - e^{-t/\tau}), \quad Q_f = 3600 \text{ C } (= 1000 \text{ mAh}), \quad \tau = 20 \text{ min} = 1200 \text{ s}.$$

The charging current is the rate at which that charge accumulates:

$$i(t) = \frac{dq}{dt} = \frac{Q_f}{\tau} e^{-t/\tau} = \frac{3600}{1200} e^{-t/\tau} = 3 e^{-t/\tau} \text{ A}.$$

So the current starts at 3 A and decays: after 20 minutes it is $3/e \approx 1.1$ A, after an hour about 0.15 A. The fuller the battery, the more slowly it fills — which is why the last ten per cent always takes so long.

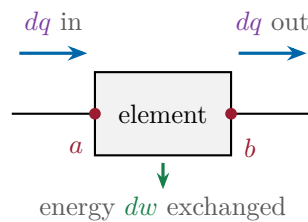
Going the other way, with a *constant* current the charge simply accumulates linearly: $I = 3\text{ A}$ flowing for 8 hours transports $Q = I \Delta t = 3 \times 8 \times 3600 = 86\,400\text{ C}$.

Charge is what accumulates; **current** is how fast it accumulates. Every integral–derivative pair in this course has that same flavour.

4. Voltage

Current tells us that charge is moving. It says nothing about *why* it moves, or what the motion costs. Charge does not wander through a circuit of its own accord: carrying it through an element always involves an exchange of energy, and voltage is the bookkeeping of that exchange.

So consider a small amount of charge dq that enters an element at terminal a and leaves it at terminal b , and let dw be the energy exchanged with the element while that happens.



Voltage (potential difference)

The **voltage** v_{ab} between terminals a and b is the energy exchanged *per unit of charge* carried from a to b through the element:

$$v_{ab} = \frac{dw}{dq}$$

The unit is the **volt (V)**, and $1\text{ V} = 1\text{ J/C}$. A positive v_{ab} means the charge *gives up* energy to the element on its way from a to b .

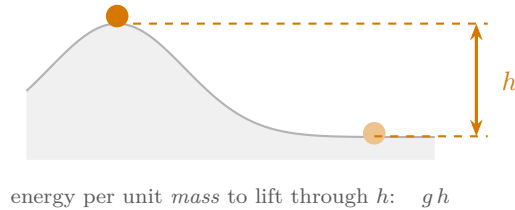
Four remarks, and they matter more than the formula.

1. **It is a ratio, and that is the whole point.** Carry twice as much charge and twice as much energy is exchanged; the ratio does not change. So v_{ab} is not a property of the charge we used to probe it — it is a property of the *pair of terminals*. (This is also why you should resist the textbook phrase “the work needed to move a test charge”: nothing has to be moved, and no particular charge is involved.)
2. **It does not require anything to be moving.** A 1.5 V battery lying in a drawer has 1.5 V across its terminals with $i = 0$, in the same way that a closed tap still has pressure behind it. Voltage is a readiness to exchange energy, not a record of energy already exchanged.
3. **It is always between two points, and the order matters:** $v_{ab} = -v_{ba}$. “The voltage at node a ” means nothing until a reference point is chosen (Section 5) — which is why the older name *potential difference* is the more honest one.
4. **In a lumped circuit, the energy per charge depends only on the two endpoints, not on the path between them.** This is not a triviality: it is assumption (b) of Section 8 in disguise, and it is what makes the phrase “the voltage between a and b ” well defined at all. Applied around a closed path, it is Kirchhoff’s voltage law.

Notice also what we get for free. If charge flows steadily, $dq = i\,dt$, so the energy exchanged per unit time is $dw/dt = v\,i$ — we will call that the power in Section 6, and it was never postulated: it follows from these two definitions.

4.1. Voltage as an electrical height

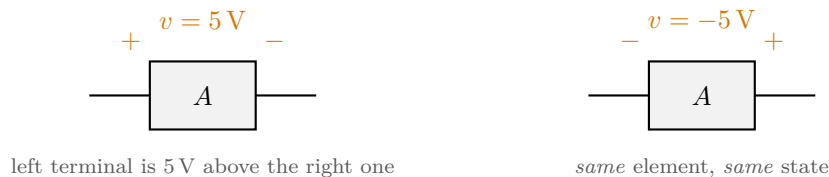
The most useful mental picture is gravitational. Lifting a mass m through a height h costs an energy mgh ; the energy *per unit mass*, gh , depends only on the height difference. A heavier ball gains more energy, but energy per kilogram is the same — so gh , exactly like voltage, is a property of the two places and not of the object you carried between them.



The picture also anticipates two later ideas. “Sea level” is a choice, and moving it changes no physical height difference — that is ground (Section 5). And the energy you spend climbing a hill does not depend on the route you take — that is the path-independence of remark 4, and it is Kirchhoff’s voltage law.

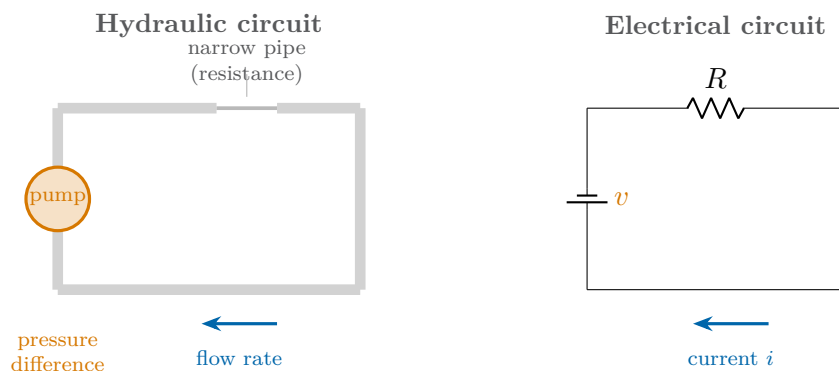
4.2. Reference polarity

Exactly as for current, we mark a *reference polarity* with a $+$ and a $-$ sign before we know the answer. The pair of pictures below describe the same physical element in the same physical state.



4.3. The hydraulic analogy

Most people have better intuition for water than for electrons, and the analogy is close enough to be genuinely useful — as long as you remember it is an analogy.



Hydraulic	Electrical	Comment
pressure (difference)	voltage v	the “across” quantity: a difference between two points
volumetric flow rate	current i	the “through” quantity: needs a direction
pipe / tap restriction	resistance R	opposes flow, dissipates energy
pump	voltage source	supplies energy
water in the pipes	charge q	conserved: it does not disappear

Where the analogy breaks

Water can leak out of a pipe and pile up in a bucket; charge in a lumped circuit cannot accumulate inside an element or a wire (Section 8). Also, “across” and “through” are not interchangeable: a sentence like “*the current across the resistor*” or “*the voltage through the resistor*” is not a slip of the tongue, it is a conceptual error, and it will cost you marks.

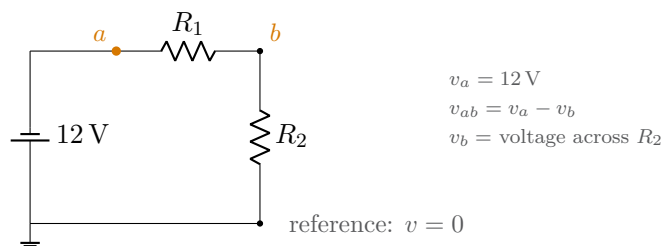
5. Ground and node voltages

Since only *differences* of potential have physical meaning, we are free to declare one point of the circuit to be zero. That point is called the **reference node**, the **datum node**, or simply **ground**.

Ground (reference node)

A node chosen, by us, to have potential zero. Once it is chosen, the “voltage at node a ”, written v_a , means $v_{a,\text{ground}}$, and every two-point voltage follows from the difference

$$v_{ab} = v_a - v_b$$



Choosing a different node as ground shifts *every* node voltage by the same constant and changes *no* two-point voltage, no current and no power. The choice is a matter of convenience: pick the node that the most elements touch, and your equations get shorter.

Three different things are called “ground”

Earth ground is a physical connection to the planet, used for safety. **Chassis ground** is the metal enclosure of an instrument. **Signal ground** is the analysis reference we have just defined. In EE201 we always mean the third one; in the laboratory, confusing them is how you blow a fuse.

6. Power and energy

Power and energy

Power is the rate at which energy is transferred:

$$p(t) = \frac{dw}{dt} = \frac{dw}{dq} \cdot \frac{dq}{dt} = v(t) i(t)$$

The unit is the **watt (W)**, $1\text{ W} = 1\text{ J/s}$. The energy transferred between t_0 and t is

$$w(t) = \int_{t_0}^t p(\tau) d\tau = \int_{t_0}^t v(\tau) i(\tau) d\tau.$$

Notice that $p = vi$ was not postulated: it fell out of the definitions of voltage (energy per charge) and current (charge per time). That is the whole point of defining them the way we did.

Units you will actually meet

Energy bills are written in kilowatt-hours: $1 \text{ kWh} = 10^3 \times 3600 = 3.6 \times 10^6 \text{ J}$. A laptop charger is 65 W; a kettle is 2000 W; the processor in your phone dissipates a few watts and that is why it gets hot.

Example 1.3 — energy from a battery

A 12 V battery supplies a constant 2 A for 5 minutes.

$$P = VI = 12 \times 2 = 24 \text{ W}, \quad W = P \Delta t = 24 \times 300 = 7200 \text{ J} = 7.2 \text{ kJ}.$$

In kilowatt-hours that is $7200/3.6 \times 10^6 = 2 \times 10^{-3} \text{ kWh}$ — which tells you something about how little energy a battery holds compared with the mains.

Example 1.5 — power that changes in time

A resistive element carries

$$v(t) = 10 \cos(\omega t) \text{ V}, \quad i(t) = 2 \cos(\omega t) \text{ A}, \quad \omega = 2\pi \cdot 50 \text{ rad/s}.$$

The instantaneous power is

$$p(t) = v(t) i(t) = 20 \cos^2(\omega t) = 10(1 + \cos 2\omega t) \text{ W}.$$

Two things are worth noticing. First, $p(t) \geq 0$ at all times: the element never returns energy, which is exactly what we expect from a resistor. Second, the power oscillates at *twice* the frequency of the voltage, around an average value of 10 W. The energy absorbed over one period $T = 1/50 \text{ s}$ is

$$w = \int_0^T p(t) dt = 10 T + \underbrace{\int_0^T 10 \cos 2\omega t dt}_{=0} = \frac{10}{50} = 0.2 \text{ J}.$$

The average power, 10 W, is what a wattmeter shows and what heats the element; the 100 Hz ripple is what makes a cheap transformer hum.

7. The passive sign convention

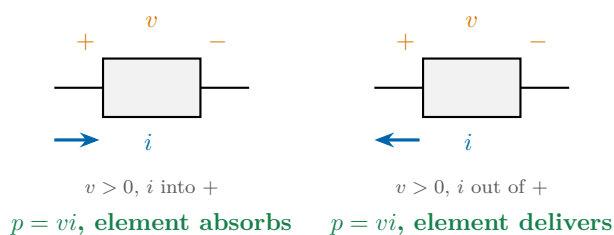
We now have $p = vi$, but v and i each carry a reference that *we* chose. Which sign of p means “this element is consuming energy”? A convention settles it once and for all.

Passive sign convention (PSC)

Assign the current reference arrow so that it **enters the terminal marked +**. With that choice,

$$p = vi = \text{power absorbed by the element}$$

so $p > 0$ means the element absorbs (consumes) energy, and $p < 0$ means it delivers (supplies) energy to the rest of the circuit.



The single most common mistake in this course

If the picture you are given does *not* satisfy the PSC — the arrow leaves the + terminal — do *not* try to remember a second formula. Flip one reference, change that variable's sign, and use $p = vi$ as always:

$$p_{\text{absorbed}} = v(-i) = -vi.$$

One convention, applied consistently, beats four memorised cases.

7.1. Conservation of power

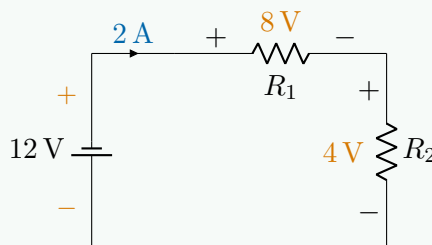
Energy is conserved, so in *any* circuit, at *every* instant, the absorbed powers of all elements sum to zero:

$$\sum_{k=1}^N p_k(t) = 0$$

Elements with $p_k < 0$ (sources, usually) supply exactly what the elements with $p_k > 0$ (resistors, usually) consume. This is Tellegen's theorem, and it is a very cheap way to check your homework: if your powers do not cancel, something is wrong.

Example 1.4 — a power balance

In the circuit below the source drives 2 A around the loop and the two resistors drop 8 V and 4 V.



With the PSC applied to each element:

$$p_{R_1} = 8 \times 2 = +16 \text{ W}, \quad p_{R_2} = 4 \times 2 = +8 \text{ W}, \quad p_{\text{source}} = 12 \times (-2) = -24 \text{ W}.$$

The source current leaves its + terminal, hence the -2 A ; the sum is $16 + 8 - 24 = 0$, as it must be. The source *delivers* 24 W and the resistors *absorb* it.

8. Electric circuits and the lumped model

Electric circuit

An **electric circuit** is an interconnection of electrical elements through which charge can flow along closed paths.

Everything we do in EE201 rests on one further modelling assumption, and it deserves to be stated precisely rather than assumed silently.

The lumped circuit model

A circuit is **lumped** when we may assume that

- (a) no net charge accumulates inside any element: the current entering an element equals the current leaving it at every instant;
- (b) no magnetic flux links any region outside the elements;
- (c) signals propagate instantaneously — the physical size d of the circuit is negligible compared with the wavelength of the highest frequency of interest,

$$d \ll \lambda = \frac{c}{f}, \quad c \approx 3 \times 10^8 \text{ m/s}.$$

When these hold, a circuit is described by *ordinary* differential equations in time. When they do not, the same physics requires Maxwell's equations and partial differential equations in space and time — which is a different, much harder course.

LUMPED

mains, $f = 50 \text{ Hz} \Rightarrow \lambda = 6000 \text{ km}$
 a 30 cm board: $d/\lambda \sim 10^{-7}$
 — circuit theory applies

NOT LUMPED

Wi-Fi, $f = 2.4 \text{ GHz} \Rightarrow \lambda = 12.5 \text{ cm}$
 a 3 cm trace: $d/\lambda \sim 1/4$
 — you need field theory

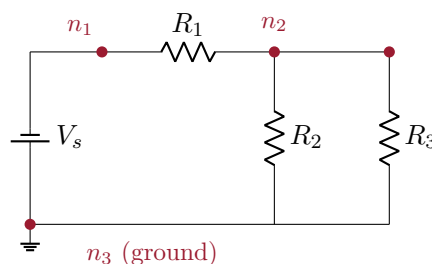
The same piece of hardware can be lumped for one signal and distributed for another. A mobile phone motherboard is a lumped circuit as far as its battery is concerned and a transmission-line problem as far as its antenna is concerned. The model is chosen by the question, not by the object.

9. Terminals, nodes, branches

The last piece of vocabulary describes how elements are connected — the topology.

Terminal, node, branch, loop

- A **terminal** is a point at which an element can be connected to others. A resistor, capacitor or inductor has two terminals; a transistor has three; an op-amp has five or more.
- A **node** is a point of connection between two or more elements. Everything joined by ideal (zero-resistance) wire is *one and the same node*, however long the wire is drawn.
- A **branch** is a single element between two nodes.
- A **loop** is a closed path through the circuit that does not pass through any node more than once.



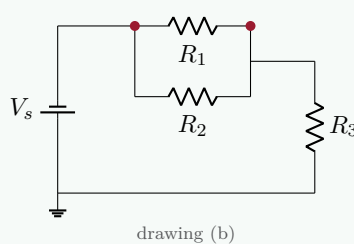
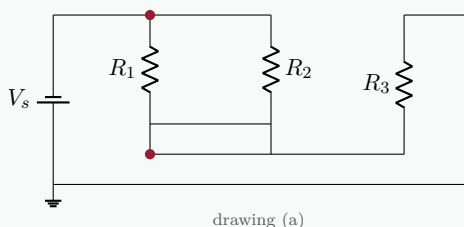
In the figure, n_2 is a single node even though three elements meet there through a stretch of wire; the wire itself is not a branch. Counting: 3 nodes, 4 branches (V_s , R_1 , R_2 , R_3) and 2 independent

loops, consistent with the relation you will meet in Lecture 2,

$$\# \text{branches} = \# \text{loops} + \# \text{nodes} - 1 \quad \implies \quad 4 = 2 + 3 - 1.$$

Example 1.6 — the same circuit, drawn twice

The two drawings below look different but are the *same* circuit: in both, R_1 and R_2 have both of their terminals in common, and that pair is in series with R_3 and the source.



Drawing (b) makes it obvious that R_1 and R_2 are in parallel. Drawing (a) hides it behind a detour of wire. Nothing was changed except where the symbols were placed — and that is the only skill being tested when a problem looks impossible at first sight.

Drawing is not topology

Two circuits that look completely different on paper can be the same circuit. What matters is which terminals share a node, not where you placed the symbols. When a problem looks unfamiliar, redraw it with the nodes lined up — half of the difficulty usually disappears.

Summary

Quantity	Symbol	Defining relation	Unit
charge	q	—	coulomb, C
current	i	$i = dq/dt$	ampere, A = C/s
voltage	v	$v = dw/dq$	volt, V = J/C
power	p	$p = dw/dt = vi$	watt, W = J/s
energy	w	$w = \int p dt$	joule, J

- Current needs a **reference arrow**; voltage needs a **reference polarity**. Both are choices. A negative answer is information, not an error.
- With the **passive sign convention** (current into the + terminal), $p = vi$ is the absorbed power, and $\sum_k p_k = 0$ over the whole circuit.
- Voltage is an **across** quantity, current is a **through** quantity; ground is the node we *choose* to call zero.
- The **lumped** assumption ($d \ll \lambda$, no charge accumulation, no stray flux) is what lets us use ordinary differential equations for the rest of the course.

Next lecture

Interconnection laws — Kirchhoff's current and voltage laws — and the branch relations of the basic elements. Everything from Lecture 1 is used there without further comment.

Reading

- Alexander & Sadiku, *Fundamentals of Electric Circuits*, Chapter 1.
- Chua, Desoer & Kuh, *Linear and Nonlinear Circuits*, Chapter 1.
- Lecture notes and videos of Prof. Emre Tuna, users.metu.edu.tr/etuna/ee201.